

RESEARCH INTENT

WQF7023 Artificial Intelligence Research Project

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Area of Research	Medical Image Segmentation / Artificial intelligence
Research Title	Nano-Mamba U-Net: A Spatio-Temporal Framework for Cardiac Motion Analysis
Introduction	Accurate cardiac motion analysis is essential for cardiovascular diagnosis. However, tracking continuous dynamic heartbeats in 4D (3D+Time) medical images is challenging. Traditional CNNs lack global receptive fields, while Vision Transformers suffer from high computational complexity, restricting their use on edge devices. This project explores the potential of State Space Models (Mamba) as a lightweight alternative to capture long-range spatio-temporal dependencies under computational constraints.
Statement of Aim	To develop a proof-of-concept “Nano-Mamba U-Net” framework and explore its feasibility for efficient and lightweight cardiac motion analysis.
Research Questions	1. How can Visual State Space Models be integrated into a lightweight U-Net architecture for spatio-temporal medical data? 2. Does the proposed Nano-Mamba framework offer a viable trade-off between segmentation accuracy and computational efficiency compared to traditional baseline models?
Research Objectives	1. To design a prototype Nano-Mamba U-Net combining CNN feature extraction with Mamba sequence modeling. 2. To implement a temporal processing mechanism for continuous cardiac frames. 3. To evaluate the strengths and limitations of the framework regarding accuracy (e.g., Dice score) and computational cost (e.g., VRAM usage).
Scope	The project focuses on 4D cardiac image segmentation using the ACDC dataset. The scope is limited to developing a prototype architecture and evaluating its performance on standard commercial GPUs (e.g., 8GB VRAM), serving as an exploratory study for edge-device deployment.
Significance of the Research	This research provides an empirical evaluation of Mamba models in dynamic medical imaging. By investigating computationally accessible AI alternatives, it contributes to the broader goal of democratizing advanced healthcare technologies (SDG 3 and SDG 9) in resource-constrained environments.

Methodology	The study will adopt an empirical approach. A lightweight CNN encoder will extract spatial features, which are then processed by Mamba bottleneck blocks to model temporal changes across the cardiac cycle. The decoder will reconstruct the segmentation maps. The model's performance will be benchmarked against a standard U-Net baseline using metrics like Dice Similarity Coefficient (DSC) and VRAM consumption to analyze its efficiency.
Expected Outcomes	1. A prototype Nano-Mamba U-Net model for 4D cardiac video processing. 2. A comprehensive evaluation report detailing the feasibility, advantages, and potential limitations of using State Space Models for dynamic medical image segmentation.
Skills Needed	Proficiency in Python, PyTorch; knowledge of CNNs and State Space Models; experience in biomedical data processing and model evaluation.
Timeline/ Gantt Chart	Phase 1 (Target Completion: June 8, 2026) • Mar - Apr: Literature review on Visual State Space Models; dataset preprocessing (ACDC). • May - Early Jun: Implementation of the standard U-Net baseline; initial design of the Nano-Mamba prototype; Phase 1 submission. Phase 2 (Target Completion: Early September, 2026) • Mid Jun - Jul: Integration of Mamba blocks and prototype training. • Aug: Comparative evaluation against the baseline; analyzing failure cases and computational trade-offs. • Sept: Finalizing results and writing the final research manuscript/thesis.